JESUS F. SANCHEZ

jesusfsa@usc.edu | jesusfsanchez.github.io | Los Angeles, CA | (805) 815-7654

EDUCATION

Master of Science - University of Southern California (Los Angeles, CA) Expected May 2026

Astronautical Engineering, Concentration: Propulsion

- Relevant Coursework:
  - Fundamentals and Applications of Combustion: Chemical thermodynamics, chemical kinetics, transport phenomena, conservation equations, non-premixed, premixed flames, and Cantera
  - Spacecraft Design: Space environment, orbital mechanics, attitude determination and control, communications, power systems, thermal control, structures and mechanisms
  - Liquid Rocket Propulsion (Spring 2025), Compressible Gas Dynamics (Spring 2025)

Bachelor of Science - California State University, Northridge (Northridge, CA) August 2024

Mechanical Engineering, Concentration: Aerospace (GPA: 3.67)

- Relevant Coursework: Rocket Propulsion, Aeropropulsion, Thermodynamics II, Fluid Dynamics (see more)

SKILLS

Ansys Fluent, Cantera, Computer-Aided Design (CAD) (Siemens NX, SOLIDWORKS), Finite Element Analysis (FEA), Geometric Dimensioning and Tolerancing (GD&T), LabVIEW, MATLAB, OpenFOAM, Python, Simulink, Teamcenter, 3D Printing

WORK EXPERIENCE

NASA Jet Propulsion Laboratory, *Mechanical Engineering Intern* May 2022 - February 2024

- Led the design of complex electromechanical systems from design to manufacturing for Mars' Sample Retrieval Lander (SRL), focusing on design for manufacturability and assembly (DFM & DFA)
- Generated engineering drawings using GD&T through Siemens NX and produced written instructions for build, assemble, and test
- Collaborated closely with technicians and the manufacturing team to support fabrication and troubleshoot hardware assembly issues, resulting in accurate and efficient production of hardware

ENGINEERING RESEARCH

Evaluation of Novel Scalable Swirl-Flame Hydrogen Combustor, *Student Research* May 2023 - Present

- Designing a novel swirl-flame combustor for gas turbine engines able to operate for hydrogen fuel
- Performing experimental analysis to study combustor capabilities and process data using MATLAB, resulting in capabilities up to 60% dilution limit
- Conducting CFD analysis through Ansys Fluent to model the flow field and combustion reaction

ENGINEERING PROJECTS

Liquid Propulsion Laboratory (LPL), *Propulsion Engineer* September 2024 - Present

- Spearheading the redesign of Mike's Fury Dev-2, a 3 kN Jet-A/LOx regeneratively cooled engine, focusing on enhanced performance, efficiency, and modularity
- Designing the thrust chamber assembly (TCA) by defining engine parameters, sizing the thrust chamber, nozzle, and injector-chamber interface
- Developing a MATLAB script utilizing the Method of Characteristics (MOC) approach for optimized rocket nozzle

ASME Human Powered Vehicle (HPV), *Fairing Team Member* August 2023 - May 2024

- Designed, analyzed (CFD & FEA), and fabricated a low-drag carbon fiber fairing with a drag coefficient of 0.43, improving from the previous team's 0.5
- Coordinated within a multidisciplinary team of 20+ students to ensure system functionality and integration, achieving 3rd place in a national competition
- Presented, verbal and written, Preliminary Design Reviews (PDR) and Critical Design Reviews (CDR) to faculty and industry professionals

NASA L'SPACE Mission Concept Academy, *Deputy Project Manager* January 2022 - March 2022

- Formulated a mission concept for a lunar rover capable of scientific exploration, evaluated by NASA experts, while leading a 10-member team across engineering, science, and business administration roles
- Conducted trade studies and established mission, vehicle, and subsystem requirements following NASA's project formulation practices

NASA MINDS Competition, *Mechanical Design Sub-Team Member* September 2020 - May 2022

- 2022 CubeSat: Conducted finite element analysis through SOLIDWORKS to optimize the design of a 4-inch cube satellite structure for weight reduction and material selection
- 2021 Autonomous Robot: Designed the drivetrain system and analyzed (FEA) the chassis of a robot intended for nondestructive testing of space vehicles, enabling it to traverse 90-degree surfaces, achieving 3rd place in competition